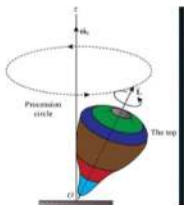


ROTATIONAL MOTION

Motion of Rigid Bodies:

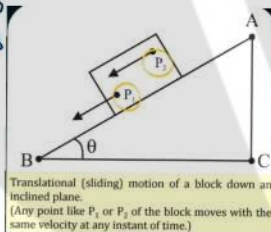
Rigid Bodies:

Rigid Body comprises of or is made up of infinite small or differential elements or particles



Pure Translational Motion:

When all the points on a rigid body moves with the same velocity, then the body is said to be in Purely Translational Motion.



* Sliding Motion is always a Translational Motion. But Translational motion is not always sliding motion.

✓ Sliding
✓ Only Translational



Sliding or Translational Motion.

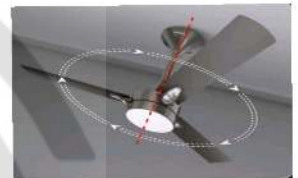


Since, there is no contact between any surface and the

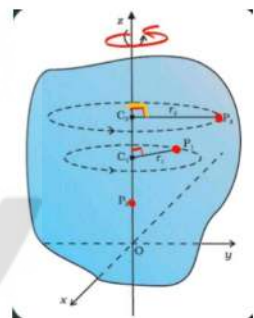
body (lift)
∴ this motion is purely Translational motion (no sliding).

Pure Rotational Motion:

When a rigid body is fixed along an axis, then all the points on the body can only rotate along that axis and can't perform translational motion, this kind of motion is said to be Purely Rotational Motion.



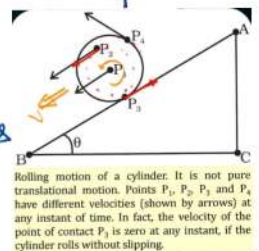
In Rotation of a rigid body about a fixed axis every particle of the body moves in a circle which lies in a plane perpendicular to the axis and has its centre on the axis.



Rotational + Translation Motion:

The velocity 'v' with which the body rotates is applied tangentially to every point on the body, thus it is responsible for rotating the body.

The velocity 'v' with which the body undergoes translational motion is same for all the points on the body.



At Point P_4 , the point is considered to be at rest, because both the

the translation velocity (v) and the tangential velocity (v') cancel out.



In Rotational + Translational Motion —

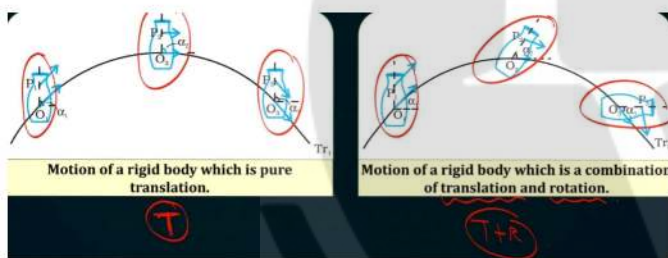
* At the Top most point / Position of the body:

The velocity is Maximum.

* At the bottom-most point / Position of the body:

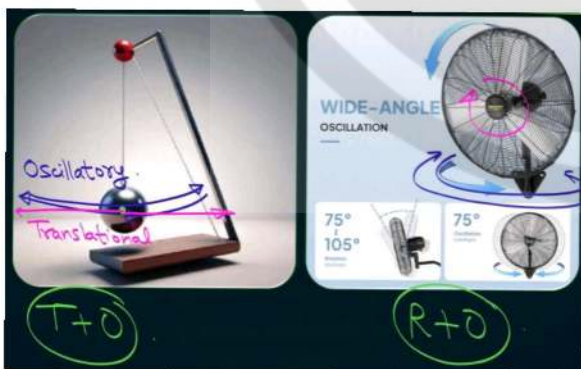
The velocity is Minimum.

⊕ Rotational + Translational Motion:



Motion of a rigid body which is pure translation.

Motion of a rigid body which is a combination of translation and rotation.



Oscillatory Translational

WIDE-ANGLE OSCILLATION

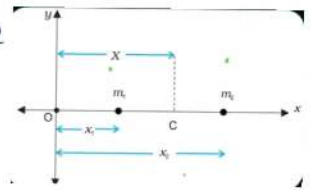


T+R+O

T+R+O

Centre of Mass:

The Centre of Mass (COM) of a system is a point that represents the average position of the entire mass of the system.



$$(X)_{\text{COM}} = \frac{m_1 x_1 + m_2 x_2}{(m_1 + m_2)} \text{ net mass of the system.}$$

$$(X)_{\text{COM}} = \frac{m_1 x_1 + m_2 x_2 + m_3 x_3 + \dots}{m_1 + m_2 + m_3 + \dots}$$

$$(X)_{\text{COM}} = \frac{\sum m_i x_i}{\sum m_i}$$

COM of a System of Particles:

If a system has 'n' particles of masses $m_1, m_2, m_3, \dots, m_n$ at position vectors $\vec{r}_1, \vec{r}_2, \vec{r}_3, \dots, \vec{r}_n$

$$\vec{R} = \frac{\sum_{i=1}^n m_i \vec{r}_i}{\sum_{i=1}^n m_i}$$

* Here, $\vec{R}$ is the position vector of the COM.

* In Cartesian form

$$x_{\text{cm}} = \frac{\sum m_i x_i}{\sum m_i}, y_{\text{cm}} = \frac{\sum m_i y_i}{\sum m_i}$$

$$z_{\text{cm}} = \frac{\sum m_i z_i}{\sum m_i}$$

⊕ COM of a Rigid Body:

co-ordinates of the COM of a system of particles.

$$X = \frac{\sum (\Delta m_i) x_i}{\sum \Delta m_i}, Y = \frac{\sum (\Delta m_i) y_i}{\sum \Delta m_i}, Z = \frac{\sum (\Delta m_i) z_i}{\sum \Delta m_i}$$

In case of a rigid body, these co-ordinates get replaced with:

$$\sum \Delta m_i \rightarrow \int dm = M,$$

$$\sum (\Delta m_i) x_i \rightarrow \int x dm$$

$$\sum (\Delta m_i) y_i \rightarrow \int y dm.$$

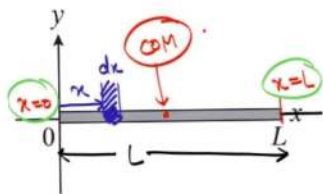
and $\sum (\Delta m_i) z_i \rightarrow \int z dm$.

Here M is the total mass of the body. The co-ordinates of the centre of mass now are -

$$x = \frac{1}{M} \int x dm$$

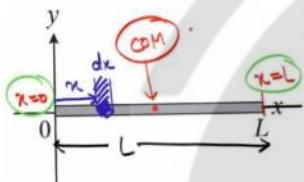
$$y = \frac{1}{M} \int y dm$$

$$z = \frac{1}{M} \int z dm$$



The vector expression equivalent to these three scalar expression is:

$$\vec{R} = \frac{1}{M} \int \vec{r} dm$$



Total mass of the rod = M

Let's consider a small portion of length dx from the rod of total length L .

Thus the mass of " dx " element will be - " dm ".

Co-ordinate of COM:

$$(x)_{\text{COM}} = \frac{\int x dm}{\int dm} = \frac{1}{M} \int x \left(\frac{M}{L} \right) dx$$

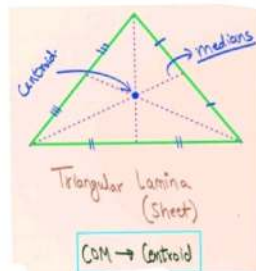
$$= \frac{1}{L} \int_0^L x dx = \frac{1}{L} \left[\frac{x^2}{2} \right]_{x=0}^{x=L}$$

$$= \frac{1}{L} \left(\frac{L^2}{2} - 0 \right) = \frac{1}{L} \left(\frac{L}{2} - 0 \right)$$

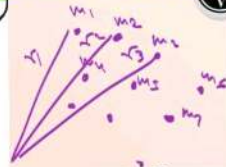
$$(x)_{\text{COM}} = \frac{L}{2}$$

Mass is linearly distributed:

$$\frac{M}{L} = \frac{dm}{dx} \Rightarrow dm = \frac{M}{L} dx$$



Motion Of Centre Of Mass:



$$M\vec{R} = m_1\vec{r}_1 + m_2\vec{r}_2 + m_3\vec{r}_3 + \dots$$

Differentiate both sides w.r.t time

$$\frac{d}{dt}(M\vec{R}) = \frac{d}{dt}(m_1\vec{r}_1) + \frac{d}{dt}(m_2\vec{r}_2) + \frac{d}{dt}(m_3\vec{r}_3) + \dots$$

$$\Rightarrow M \frac{d\vec{R}}{dt} = m_1 \frac{d\vec{r}_1}{dt} + m_2 \frac{d\vec{r}_2}{dt} + m_3 \frac{d\vec{r}_3}{dt} + \dots$$

$$\Rightarrow M\vec{V} = m_1\vec{v}_1 + m_2\vec{v}_2 + m_3\vec{v}_3 + \dots$$

→ If the system of particles in which $m_1, m_2, m_3, \dots$ are moving with velocities $v_1, v_2, v_3, \dots$ respectively then the velocity of COM of the particles will be ' V '.

$$M\vec{V} = m_1\vec{v}_1 + m_2\vec{v}_2 + m_3\vec{v}_3 + \dots$$

Differentiating both sides w.r.t time:

$$\Rightarrow M \frac{d\vec{V}}{dt} = m_1 \frac{d\vec{v}_1}{dt} + m_2 \frac{d\vec{v}_2}{dt} + m_3 \frac{d\vec{v}_3}{dt} + \dots$$

$$\Rightarrow M\vec{a} = m_1\vec{a}_1 + m_2\vec{a}_2 + m_3\vec{a}_3 + \dots$$

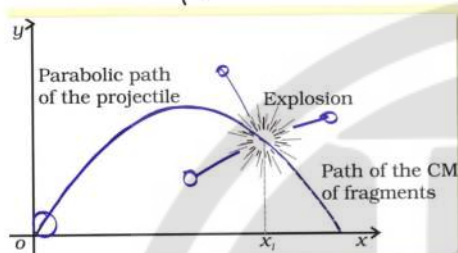
→ If we have a system of particles of mass $m_1, m_2, m_3, \dots$ having the acceleration $a_1, a_2, a_3, \dots$ then $\vec{a}$ is the acceleration of the COM of the system of particles.

$$\Rightarrow M\vec{a} = \vec{F}_{\text{external}}$$

Motion of Centre of Mass:

$$MA = F_{ext}$$

Where F_{ext} represents the sum of all external forces acting on the particles of the system. Equation states that the entire centre of mass of the system of particles moves as if all the mass of the system was concentrated at the centre of mass and all the external forces were applied at that point.



The centre of mass of the fragments of the projectile continues along the same parabolic path which it would have followed if there were no explosion.

Vector Product of Two Vectors:

→ CROSS PRODUCT

$$\vec{A} = 2\hat{i} + 3\hat{j} + 2\hat{k}$$

$$\vec{B} = \hat{i} + 2\hat{j} + 4\hat{k}$$

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 2 \\ 1 & 2 & 4 \end{vmatrix}$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 2 \\ 1 & 2 & 4 \end{vmatrix}$$

$$= \hat{i}(3 \times 4 - 2 \times 2) - \hat{j}(2 \times 4 - 1 \times 2) + \hat{k}(2 \times 2 - 1 \times 3)$$

$$= \hat{i}(8) - \hat{j}(6) + \hat{k}(1)$$

$$\vec{A} \times \vec{B} = 8\hat{i} - 6\hat{j} + \hat{k}$$



Rule of the right hand for defining the direction of the vector product.



Relation Between Angular Velocity and Linear Velocity:

Angular Velocity $\rightarrow \omega$

Linear Velocity $\rightarrow v$

$$v = \omega r$$

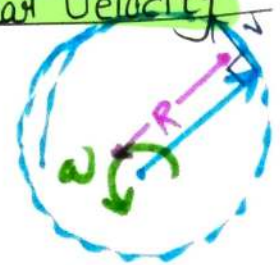
Sense of Angular velocity:

→ clockwise

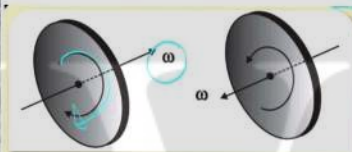
→ Anticlockwise

$$\vec{v} = \vec{\omega} \times \vec{r}$$

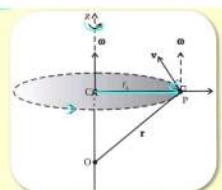
Linear velocity of a particle undergoing rotation will be the cross product of the Angular velocity vector and radius vector.



Right Hand Rule:



If the head of a right-handed screw rotates with the body, the screw advances in the direction of the angular velocity ω . If the sense (clockwise or anticlockwise) of rotation of the body changes, so does the direction of ω .



The angular velocity vector ω is directed along the fixed axis as shown. The linear velocity of the particle at P is $v = \omega \times r$. It is perpendicular to both ω and r and is directed along the tangent to the circle described by the particle.

Angular Acceleration:

⊕ Linear -

Displacement $\rightarrow \vec{R}$

Velocity $\rightarrow \frac{d\vec{R}}{dt} = \vec{v}$

Acceleration $\rightarrow \frac{d\vec{v}}{dt} = \frac{d^2\vec{R}}{dt^2} = \vec{a}$

⊕ Circular / Rotational -

Angular Displacement $= \vec{\theta}$ (rad)

Angular Velocity $\rightarrow \frac{d\vec{\theta}}{dt} = \vec{\omega}$ (rad/s)

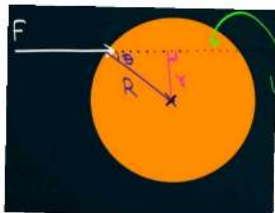
Angular Accⁿ $\rightarrow \frac{d\vec{\omega}}{dt} = \frac{d^2\vec{\theta}}{dt^2} = \alpha$
 (rad/s²)

Moment Of Force (Torque):

Torque is the turning effect produced by a force when it is applied at some distance from the axis of rotation (or Pivot point). It is also called Moment of Force or Torque or couple.

$$\sin\theta = \frac{r}{R} \Rightarrow r = R\sin\theta$$

$$R\sin\theta = r$$



$$\vec{\tau} = \vec{r} \times \vec{F}$$

S.I Unit of Torque = Nm.
 r = Perpendicular distance.

Angular Momentum Of Particle:

Angular Momentum: $\vec{L} = \vec{r} \times \vec{P}$

$$\vec{v} = \vec{\omega} \times \vec{r}$$

Angular velocity

Differentiating it both sides w.r.t time

$$\frac{d\vec{L}}{dt} = \frac{d}{dt}(\vec{r} \times \vec{P}) = \left(\frac{d\vec{r}}{dt}\right) \times \vec{P} + \vec{r} \times \left(\frac{d\vec{P}}{dt}\right)$$

$$\frac{d\vec{L}}{dt} = \vec{v} \times m(\vec{v}) + \vec{r} \times \vec{F}$$

$$= |\vec{v}|m|\vec{v}| \sin\theta + \vec{\tau}$$

$$\Rightarrow \frac{d\vec{L}}{dt} = \vec{\tau}$$

$$\begin{cases} J = \Delta P = F \times \Delta t \\ \Rightarrow F = \frac{\Delta P}{\Delta t} \\ \Rightarrow \vec{F} = \frac{d\vec{P}}{dt} \end{cases}$$

Torque And Angular Momentum for a System of Particles:

$$\vec{L} = \vec{r} \times \vec{P}$$

- Vector Quantity
- SI Unit: kg m² s⁻¹
- Dimensional Formula: [M L² T⁻¹]

$$\vec{\tau} = \vec{\tau}_1 + \vec{\tau}_2 + \vec{\tau}_3 + \dots$$

$$\vec{L} = \vec{L}_1 + \vec{L}_2 + \vec{L}_3 + \dots$$

Conservation of Angular Momentum

When we have a system of particles, where there is no external force or Torque acting on it from outside the system, then we can say that initial Angular Momentum & final Angular Momentum are equal, therefore the Angular Momentum is conserved or remains constant.

$$\vec{L} = \text{constant}$$

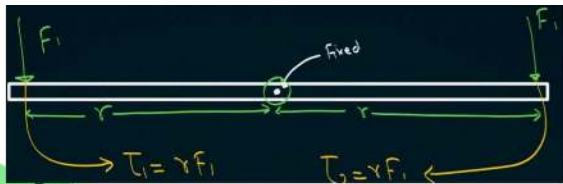
$$\Rightarrow \vec{L}_i = \vec{L}_f$$

Equilibrium Of Rigid Bodies:



A Rigid body is said to be in Eq^m when:

- ① It has no Translational Motion (does not move from one place to another)
 - ② It has no rotational motion (does not rotate about any axis).
- * In simple words $\rightarrow$ A Rigid body in Eq^m remains at rest or continues to move with uniform velocity without rotation.



Conditions for Equilibrium of a Rigid Body:

1. Translational Equilibrium

- The vector sum of all external forces acting on the body must be zero.

$$\sum \vec{F} = 0$$

- This ensures no linear acceleration of the body.

2. Rotational Equilibrium

- The Algebraic sum of all torques acting on the body must be zero about any axis.

$$\sum \vec{\tau} = 0$$

- This ensures no angular accⁿ of the body.

If the Rigid Body is fixed or Pivoted along a Point/axis:

Then we will conserve its Torque about that fixed point.

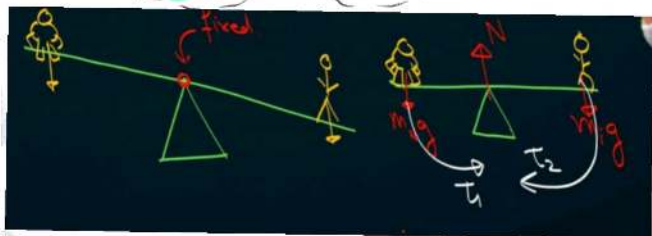
$$\sum \vec{\tau} = 0$$

If the body is not fixed, and is in Equilibrium:

Then we will conserve forces acting on it for Translational Motion and net Torque will be also conserved about the point where maximum no. of forces are acting.

$$\sum \vec{F} = 0$$

$$\sum \vec{\tau} = 0$$



DADA Method:

If the body is not fixed at any point or any axis, then we will conserve the Torque about the point where maximum no. of forces are acting.

Centre Of Gravity:

- The Centre of Gravity of a body is the point through which the entire weight of the body appears to act, irrespective of the position of the body.

- In other words, it is the point where the resultant weight of all the particles of the body acts.

Key Points:

- The C.G. may lie inside or outside the body.
- Example: Inside → a solid Sphere, Cube.
- Outside → a ring, hollow sphere, or a boomerang.
- Every rigid body has a unique centre of gravity.
- The Position of the C.G. depends on the shape and mass distribution of the body.
- For Symmetric bodies of uniform density, the C.G. lies at the geometric centre.

Moment of Inertia:

Definition:

The Rotational analogue of mass in Linear motion.

It Measures the resistance of a body to change its rotational motion about a given axis is called the Moment of Inertia.

of that Body about that given axis.

Expression:

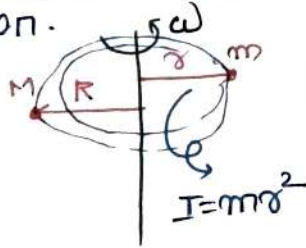
For a Particle of mass 'm' at a Perpendicular distance 'r' from the axis of rotation.

$$I = mr^2$$

For a system of Particles:

$$I = mr^2 + MR^2$$

$$I = \sum m_i r_i^2$$



→ Analogy with Linear Motion:

- Linear Motion → Mass (m) resists acceleration.
- Rotational Motion → Moment of Inertia (I) resists angular acceleration.

* Factors affecting M.I.:

- Mass of the Body (m) - Greater the mass, greater the M.I.
- Distribution of mass - farther the mass from the axis, greater the M.I.
- Axis of Rotation - Orientation of the axis changes the M.I.

Units & Dimensions:

- SI Unit: kg-m^2
- Dimension: $[ML^2]$



Linear Motion

↳ Mass $\Rightarrow m$

↳ Velocity $\Rightarrow v$

$$K = \frac{1}{2} mv^2$$

$$v = \omega r$$

$$\Rightarrow (K.E)_{\text{Rot}} = \frac{1}{2} m \omega^2 r^2$$

$$= \frac{1}{2} (mr^2) \omega^2$$

$$= \frac{1}{2} I \omega^2$$

Rotational Motion

↳ MOT $\Rightarrow I$

↳ Angular Velocity $\Rightarrow \omega$

$$\text{Rot. KE} = \frac{1}{2} I \omega^2$$

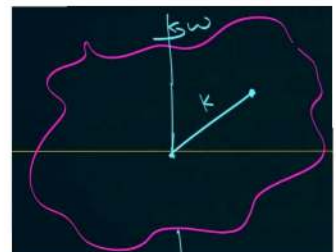
Moment of Inertia of Rigid Bodies

S	Body	Axis	Figure	I
(1)	Thin circular ring, radius R	Perpendicular to plane, at centre		MR^2
(2)	Thin circular ring, radius R	Diameter		$MR^2/2$
(3)	Thin rod, length L	Perpendicular to rod, at mid point		$ML^2/12$
(4)	Circular disc, radius R	Perpendicular to disc at centre		$MR^2/2$
(5)	Circular disc, radius R	Diameter		$MR^2/4$
(6)	Hollow cylinder, radius R	Axis of cylinder		MR^2
(7)	Solid cylinder, radius R	Axis of cylinder		$MR^2/2$
(8)	Solid sphere, radius R	Diameter		$2MR^2/5$

Radius of Gyration (K):

The distance from the axis of rotation at which the whole mass of the body remains or can be assumed to be concentrated so that its Moment of Inertia remains the same.

$$K = \sqrt{\frac{I}{M}}$$



Kinematics of Rotational Motion about a fixed axis:

Equations of Linear Motion:

- * $v = u + at$
- * $s = ut + \frac{1}{2}at^2$
- * $2as = v^2 - u^2$

- v = final velocity
- u = initial velocity
- t = time
- s = Linear displacement
- a = Linear acceleration

v	$\rightarrow$	ω
u	$\rightarrow$	ω_0
t	$\rightarrow$	t
s	$\rightarrow$	θ
a	$\rightarrow$	α

Equations of Rotational Motion:

- * $\omega = \omega_0 + \alpha t$
- * $\theta = \omega_0 t + \frac{1}{2}\alpha t^2$
- * $2\alpha\theta = \omega^2 - \omega_0^2$

- ω = final Angular Velocity (rad/s)
- ω_0 = initial Angular Velocity (rad/s)
- t = time
- θ = Angular displacement (rad)
- α = Angular acceleration (rad/s²)